Challenge: ML-based downscaling of future climate scenarios

Mikhail Ivanov and Ramón Fuentes-Franco

Rossby Centre, SMHI

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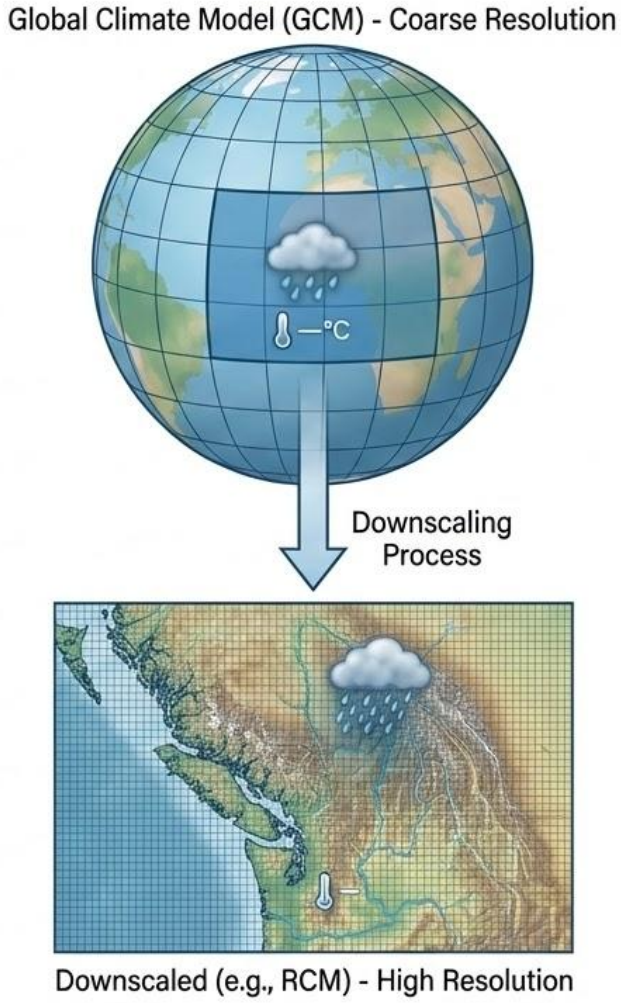
National Technical University of Athens

Downscaling

Why do we need downscaling?

Global Climate Models (GCMs) lack regional detail due to their coarse spatial resolution. Many climate applications therefore require high-resolution information derived from GCM output.

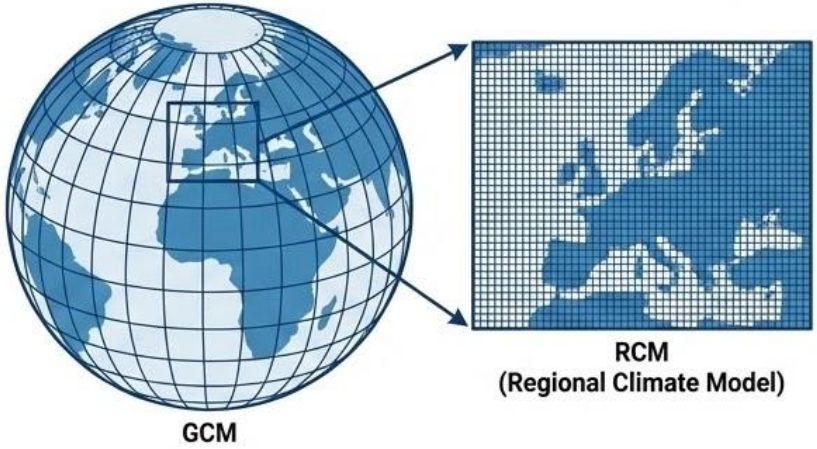
- Regional climate emerges from interactions between large-scale atmospheric circulation and local physiographic features (e.g., topography, coastlines, land–atmosphere interactions).
- Many important processes occur at spatial scales unresolved by GCMs, including mesoscale dynamics and local convection.
- There is a growing need to better understand the processes controlling regional climate variability and extremes.
- Impact assessments and adaptation planning require high-resolution climate information to evaluate vulnerability and risks.



Downscaling

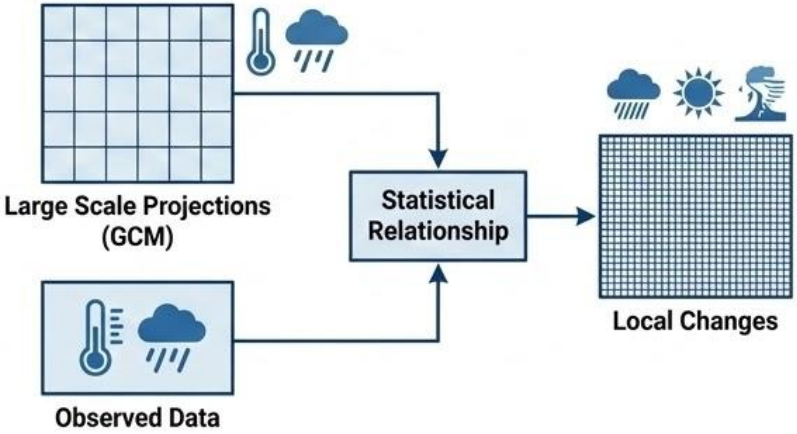
Dynamical Downscaling

This is essentially running models that are similar to GCMs, but for specific regions. Because these Regional Climate Models (RCMs) cover a smaller area, they can have higher resolution than GCMs and still run in a reasonable time.



Statistical Downscaling

This involves using observed data to establish a statistical relationship between the global and local climate. Using this relationship, scientists then derive local changes based on the large scale projections coming from GCMs or observations.



Dynamical downscaling

GCM

Grid boxes

~100-400 km wide

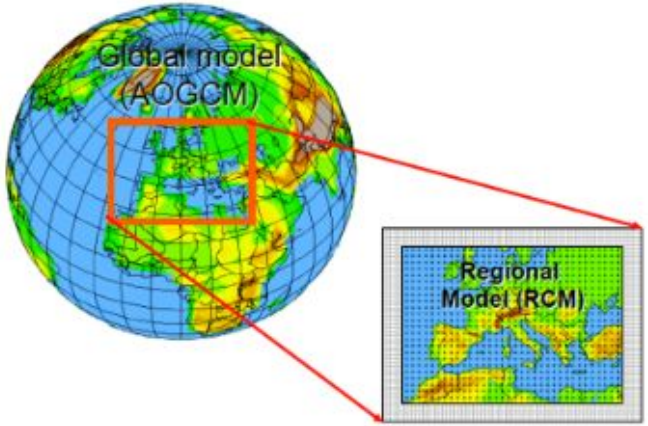


Fig. 1: Regional climate downscaling provides greater detail at local and regional scales, which is vital to improving analyses of vulnerability, impact and adaptation

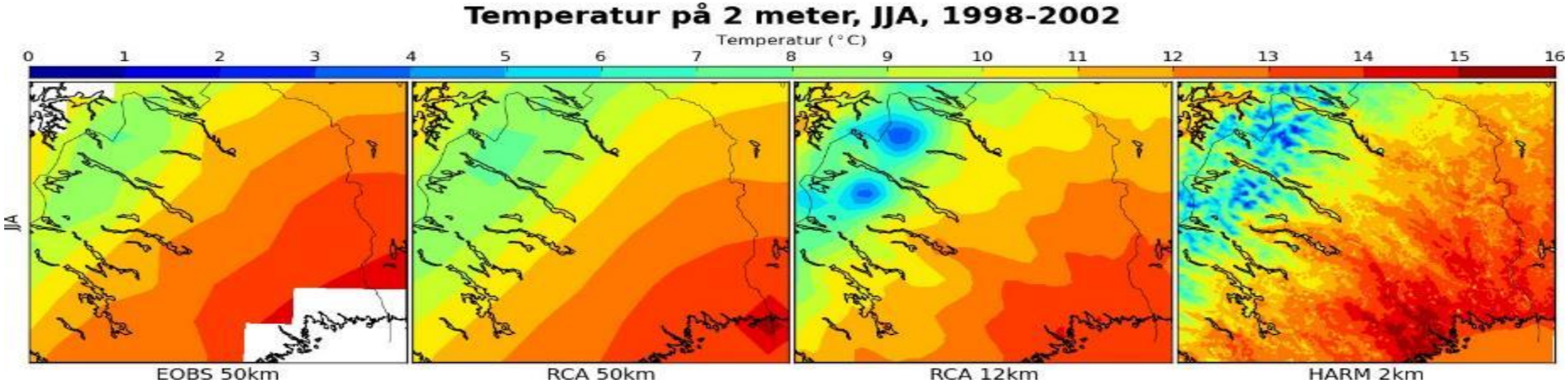
(Illustration courtesy of Justin Glisan)

RCM

Grid boxes

~ 10-50 km wide

Dynamical downscaling

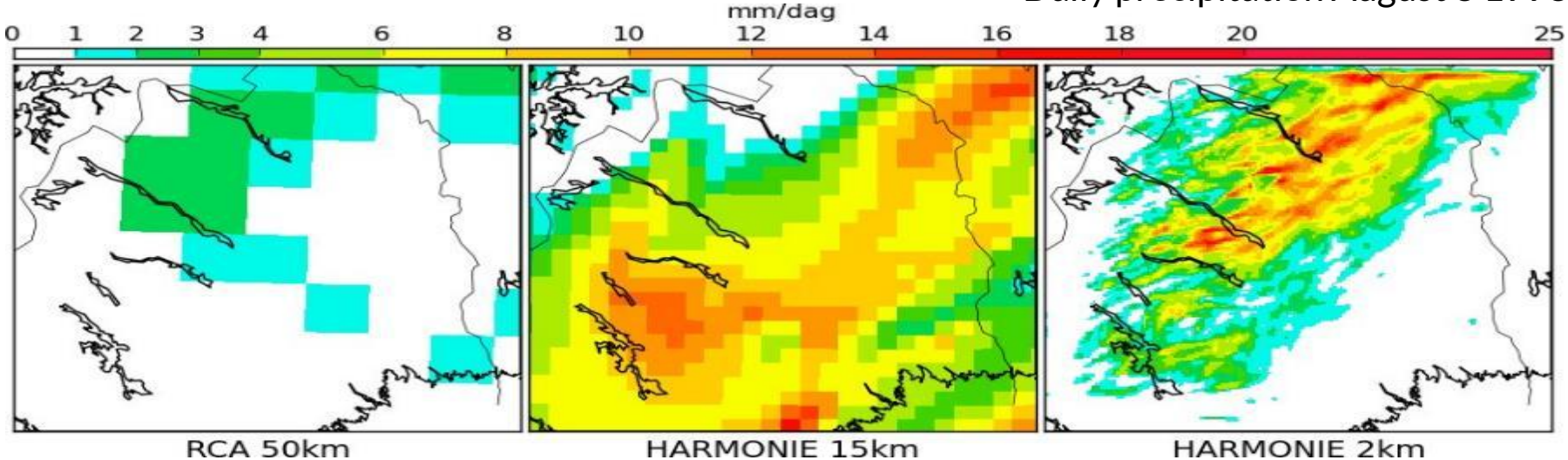


You get more and more detail…

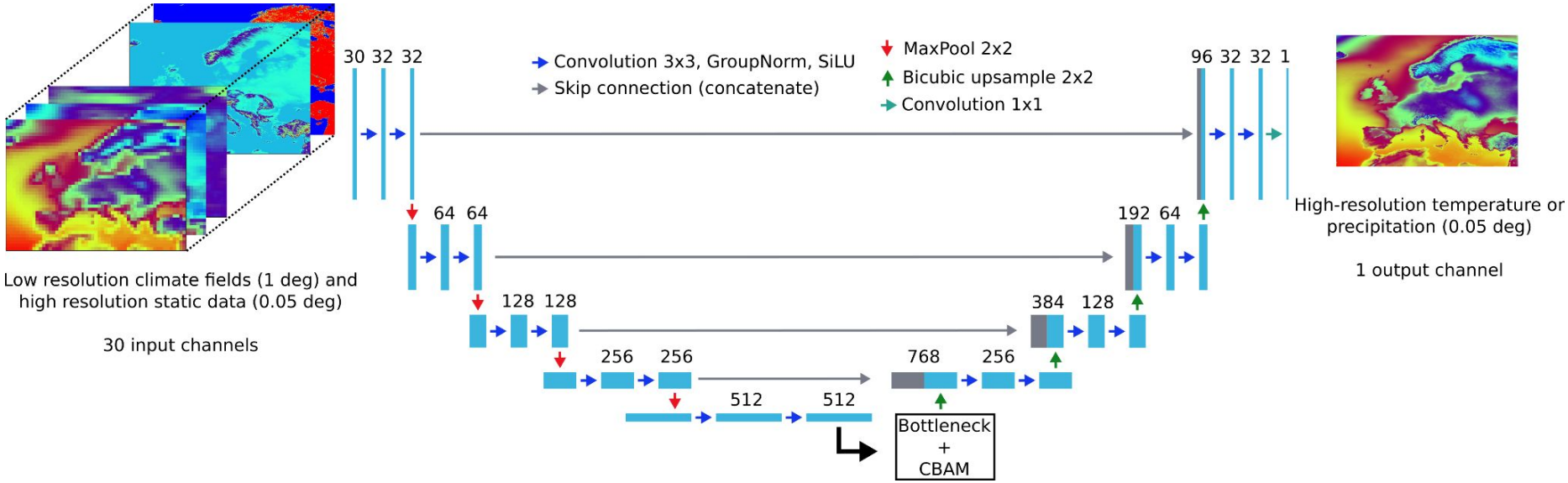
Dynamical downscaling

If precipitation is an important parameter in your study, downscaling from Global Climate Model (GCM) scale is usually needed. Due to the higher horizontal resolution, Regional Climate Models (RCMs) can better represent horizontal heterogeneity in precipitation patterns as induced by complex topography and/or land/sea contrasts.

Daily precipitation August 8 1998



ML architecture diagram (UNet)

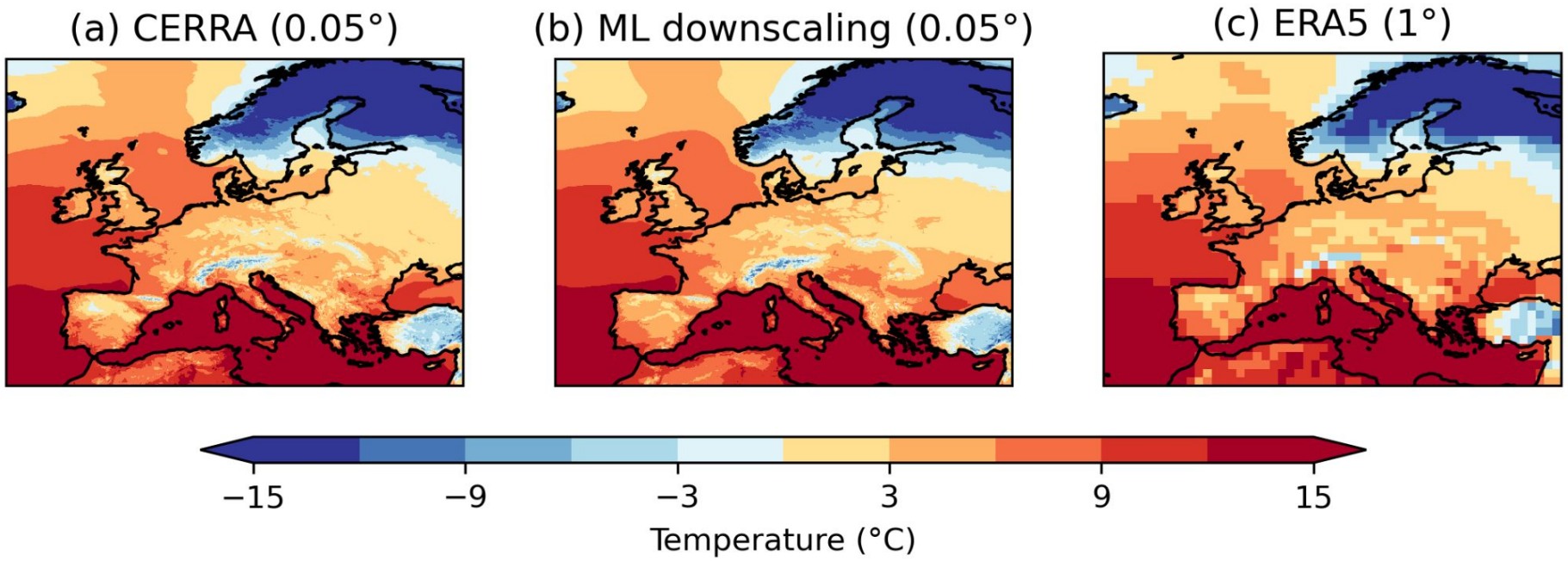


[1] Ramón Fuentes Franco et al., "Pan-European high-resolution downscaling using deep learning", JGR ML and Comput., 2025 (accepted)

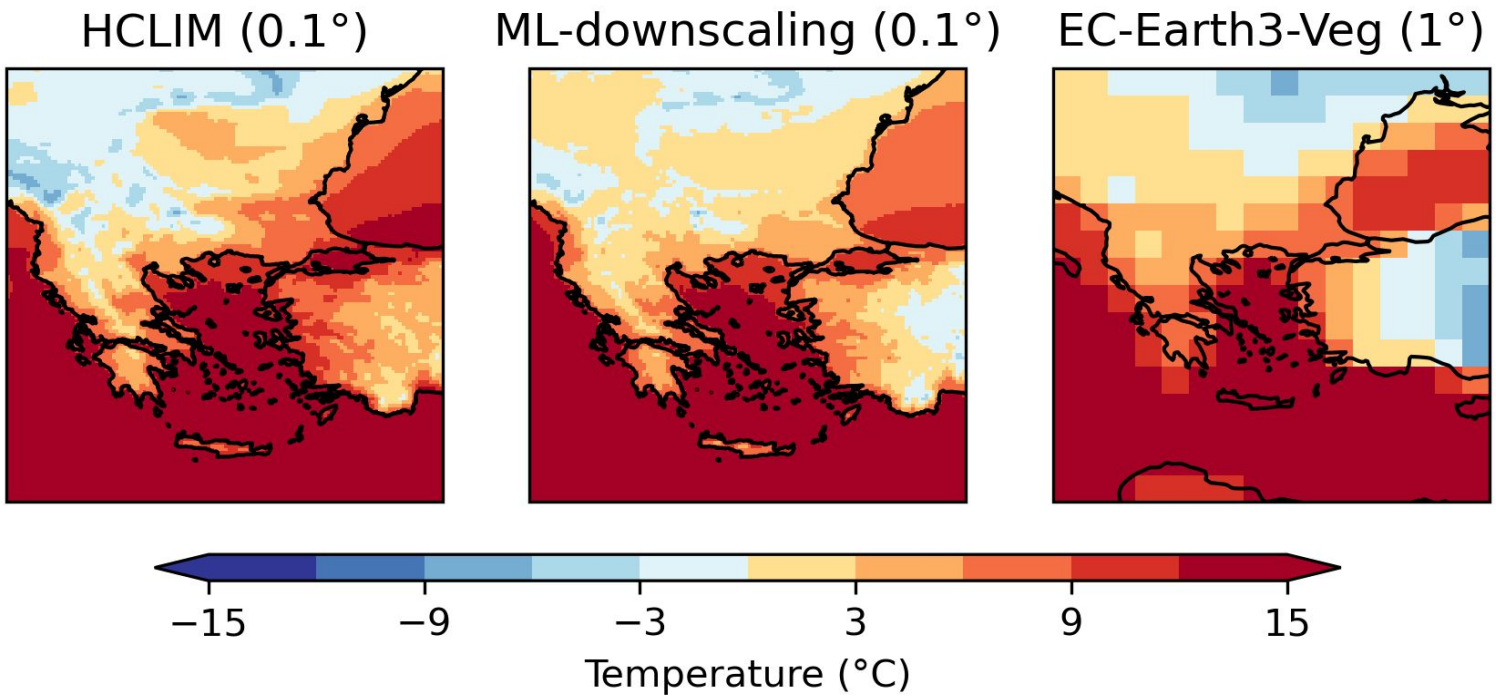
[2] Olaf Ronneberger et al., "U-Net: Convolutional Networks for Biomedical Image Segmentation", MICCAI, 2015, pp. 234-241

[3] Sanghyun Woo et al., "CBAM: Convolutional Block Attention Module", ECCV, 2018, pp. 3-19

ML downscaling example



ML downscaling example (challenge subdomain)



Example of research questions for the challenge

- Training on model data vs reanalysis (observational) data
- Heatwave analysis
- Spatial transferability (e.g. test a trained model in a different subdomain)
- Effect of training dataset extension
- Extreme event detection
- Your own idea?

To begin your work

- Go to the challenge repository on GitHub: https://github.com/WinterSchool2026/ch02-future-climate-downscaling
- Download the starter notebook: `ch02_future_climate_downscaling.ipynb`
- Upload the notebook to your Google Drive (you need a google account for that) and open it using Google Colab service (https://colab.research.google.com/)